# PRODUCT USER MANUAL

## For
## OMI_CLIMATE_SI_BALTIC_volume

**Issue: 4.0**

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# RECORD TABLE

| Issue | Date | § | Description of Change | Author | Validated By |
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# GLOSSARY AND ABBREVIATIONS

| | |
|---|---|
| **MOI** | Mercator Ocean International |
| **NetCDF** | Network Common Data Form |
| **OMI** | Ocean Monitoring Indicator |
| **PUM** | Product User Manual |
| **QUID** | Quality Information Document |
| **SIV** | Sea Ice Volume |

# DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

# 1) INTRODUCTION

## a) Summary

This document is the user manual for the Copernicus Marine Service Ocean Monitoring Indicator (OMI) OMI_CLIMATE_SI_BALTIC_volume. It contains two datasets:

- omi_climate_si_baltic_volume
- omi_climate_si_baltic_volume_yday

The OMI is derived from daily sea ice concentration products SEAICE_BAL_PHY_L4_MY_011_019 and SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004. The data is daily time series of the ice volume in the Baltic Sea derived from the operational ice charts.

Ice charts are produced by manually drawing polygons of ice, where each polygon has a set of attributes, including sea ice concentration and level ice thickness. The data used to produce the charts are satellite images (SAR) as well as in situ observations (reports) from ice breakers in the Baltic Sea. The ice charts are produced almost daily during the ice season and cover the period November 1979 to present.

Daily SIV values are computed from the daily sea ice thickness and concentration maps. All sea ice covered ocean is included, lake ice is not. SIV is computed as the sum of the area of the grid cells that have a SIC value larger than 15% multiplied with corresponding sea ice thickness.

The product is published on the Copernicus Marine Service dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the Copernicus Marine Service Information System. Files downloaded are in NetCDF format and follow CF-1.7 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is further described in the corresponding Quality Information Document (QUID).

# 2) DESCRIPTION OF THE PRODUCT SPECIFICATION

## a) General Information

| | |
|---|---|
| **Product Lines** | OMI_CLIMATE_SI_BALTIC_volume |
| **Geographical coverage** | Baltic Sea |
| **Variables** | Sea ice volume |
| **Available time series** | From 1993 to Month-4 |
| **Temporal resolution** | Daily |
| **Horizontal resolution** | Integrated |
| **Integration depth (for example)** | surface |
| **Format** | NetCDF4 |

## b) Production System Description

The sea ice volume (SIV in km3) time series have been derived from daily sea ice charts from SEAICE_BAL_PHY_L4_MY_011_019 and SEAICE_BAL_SEAICE_L4_NRT_OBSERVATIONS_011_004. These products provide sea ice concentration and thickness for the Baltic Sea from January 1980 until present. Weekly smoothed time series of the SIV have been calculated from daily values using moving average filter with 7-day window.

The annual course of the SIV has been calculated as daily mean ice volume for each day-of-year in the period of 1993-present. Same way the standard deviation is calculated from ice areas occurring during each of the day-of-year.

The SIV from the time period of October to September is used to show ice volume during the last winter.

## c) Dataset details

| |
|---|
| **omi_climate_si_baltic_volume** (daily sea ice volume time series smoothed over 7-day time period) contains the following variables: |
| **sivoln** [10e3 km3]<br>long_name: Sea Ice Volume<br>standard_name: sea_ice_volume<br>**time** [days since 1950-01-01 00:00:00]<br>long_name: time<br>standard_name: time |
| **omi_climate_si_baltic_volume_yday** (time series of mean year day sea ice volume) contains the following variables: |
| **sivoln_yday_mean** [10e3 km3]<br>long_name: Year Day Average Sea Ice Volume<br>standard_name: sea_ice_volume<br>**sivoln_yday_std** [10e3 km3]<br>long_name: Year Day Sea Ice Volume Standard Deviation<br>standard_name*: sea_ice_volume<br>**sivoln_yday_yyyy** [10e3 km3]<br>long_name: Sea Ice Volume of yyyy-1/yyyy<br>standard_name*: sea_ice_volume<br>**time** [days since 1950-01-01 00:00:00]<br>long_name: time<br>standard_name: time |

* The standard name is defined closest available in CF conventions yet it does not reflect the content of the variable. Please use variable name as a reference to the content.

# 3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

# 4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Files nomenclature when using the get command in the Copernicus Marine Toolbox or using the Browse File is as follows:

<datasetname>_P<productiondate>.nc

Where :
<datasetname> is the dataset reference
<productiondate> is the production date when the file was generated
.nc is the standard NetCDF filename extension.

# 5) REFERENCES

Quality Information Document (CMEMS-BAL-OMI-QUID-CLIMATE-SI-volume)